Highlights

Optimization of a Recycling Mixer-Reactor-Clarifier Activated Sludge System Using a Physically-Constrained Statistical Surrogate

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- The formulation admits arbitrary reactor and Clarifier-layer counts.
- Stage residence time and aeration can be manipulated independently.
- A unique convex projection reconciles the complete surrogate response.
- A primal–dual gap yields a smaller single-level optimization model.
- Reference nonsmooth replay is used to validate every selected decision.

Optimization of a Recycling Mixer-Reactor-Clarifier Activated Sludge System Using a Physically-Constrained Statistical Surrogate

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ABSTRACT

Activated-sludge optimization is difficult because biological conversion, internal mixed-liquor recycle, return activated sludge, wasting, and secondary clarification form one nonlinear closed loop. This work formulates the system for an arbitrary number N of completely mixed reactors and an arbitrary number L of Clarifier layers. The freely varied coordinates may include stage residence times, independent stage aeration settings, recycle ratios, wasting, and other explicitly balanced stage inputs. A quadratic multiresponse surrogate predicts the mixer, every reactor outlet, both Clarifier outlet component-flow vectors, and the layer-solids profile. Its $n_\chi = (N+3)F+L$ responses are reconciled by a unique scaled-Euclidean projection that imposes recycle mixing, stoichiometric-invariant transport, Clarifier component conservation, soluble pass-through, particulate densification, endpoint consistency, an admissible layer envelope, and non-negativity. Operational optimization is constrained by this projected state. A primal–dual-gap reformulation expresses the lower projection without complementarity slacks and is independently verified by resolving the original quadratic program. The case study uses five equal-volume reactors, independently adjustable aeration in reactors 3–5, a ten-layer Clarifier, 1,000 mechanistic design states, five-fold cross-validation, and ten influent scenarios. Raw, projected, smooth-mechanistic, and reference nonsmooth responses will be compared stage by stage. The formulation separates regression error, physical reconciliation, decision quality, smoothing error, and computational cost without asserting global optimality.

Nomenclature

Only nonstandard mathematical symbols are listed. All vector inequalities are componentwise.

$A \in \mathbb{R}^{K\times F}$	Full-row-rank stoichiometric-invariant operator
A_{cl}	Clarifier surface area
a_i, $k_L a_i$	Normalized aeration setting and oxygen-transfer coefficient in reactor i
$c_i, m, c_E, c_U \in \mathbb{R}^F_+$	Reactor-i, mixer, Clarifier-overflow, and Clarifier-underflow concentrations
D_χ, D_y, D_b, D_g, D_T	Positive response, mechanistic-state, physical-row, and quality scales
d_i	Recycle-adjusted dilution rate in reactor i, d^{-1}
E_S, E_X	Selectors for soluble and particulate components
F, F_S, F_X	Total, soluble, and particulate component counts
$g_E = q_E c_E$, $g_U = q_U c_U$	Clarifier outlet component flows divided by fresh flow
h_i, H	Nominal stage and total HRT on a fresh-flow basis, h
I_{comp}	Component-to-COD/TN/TP/TSS reporting map
J	Dimensionless engineering objective
K	Number of independent stoichiometric invariants
L, ℓ_F	Number of Clarifier layers and feed-layer index
N, n_p	Numbers of completely mixed biological stages and kinetic processes
n_χ, n_y, n_ϕ, d_z	Complete-response, mechanistic-state, feature, and statistical-input dimensions
q_P, q_C, q_E, q_U	Reactor-train, Clarifier-feed, overflow, and underflow flow ratios
Q_0	Fresh-influent flow
r_I, r_R, w	MLR, RAS, and WAS ratios relative to Q_0

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s_ℓ	TSS concentration in Clarifier layer ℓ
t_X	Component-to-TSS column vector
$\vartheta \in \mathbb{R}^{d_\vartheta}$	Vector containing exactly the freely varied operating coordinates
$x \in \mathbb{R}_+^F$	Fresh-influent component vector
$\chi_{\text{raw}}, \hat{\chi}$	Raw-surrogate and physically projected complete responses
y	Reactor-and-Clarifier mechanistic state
ν, ρ	Stoichiometric matrix and process-rate vector
ϕ	Standardized unique-monomial feature vector
ϖ, ϖ_Q	Overall objective priorities and effluent-quality priorities

1. Introduction

An activated-sludge plant with mixed-liquor recycle (MLR), return activated sludge (RAS), waste activated sludge (WAS), and a secondary Clarifier is one recycling process. Changing an internal recycle alters reactor throughflow; changing aeration alters both conversion and power demand; changing wasting alters solids age; and the Clarifier determines the treated-water quality and the solids returned to the biology. Treating these decisions as independent subsystems would miss the loop that makes the plant operate.

Activated Sludge Models describe this coupling mechanistically, while multilayer settling models represent clarification and thickening (Henze, Gujer, Mino, and van Loosdrecht, 2006; Takács, Patry, and Nolasco, 1991). Repeatedly integrating those equations to a stable steady state is expensive inside operational optimization. A statistical surrogate can be evaluated much faster, but an unconstrained regression may predict a negative concentration, lose a conserved inventory between reactors, or create component mass in the Clarifier. Optimization amplifies this risk because its purpose is to search for unusually favorable predictions (Bhosekar and Ierapetritou, 2018).

The present formulation gives regression and physics different, auditable roles, extending invariant-aware surrogate modeling from isolated responses to the complete recycling plant (Selerio, 2026). A quadratic multiresponse model predicts the closed-loop response. A strictly convex quadratic program (QP) then finds the nearest response satisfying a stated set of linear physical relations. The optimizer uses only this projected response. Nonlinear biological rates and the nonlinear settling law are not attributed to the projection; they are checked independently with the mechanistic model. This distinction is important because satisfying conservation is necessary for a credible plant state, but it is not sufficient to establish kinetic accuracy.

The derivation is dimension-parametric. It applies to N biological stages, L Clarifier layers, and any declared set of free stage and plant controls. Numerical values—including N, L, the aerated stages, operating bounds, objective priorities, and engineering limits—are introduced only in the case study. The case uses five reactors and ten layers, but neither number is built into the theory.

2. Theory and calculation

2.1. Walking once around the recycle loop

Consider $N \geq 1$ completely mixed biological stages followed by an $L \geq 3$ layer Clarifier. Component vectors contain $F = F_S + F_X$ entries; $E_S \in \mathbb{R}^{F_S \times F}$ and $E_X \in \mathbb{R}^{F_X \times F}$ select a disjoint partition of the soluble and particulate entries, so $E_S^\top E_S + E_X^\top E_X = I_F$ and $E_S E_X^\top = 0$. The non-negative TSS map t_X has zero soluble entries. Figure 1 follows one unit of fresh influent around the plant. The final reactor supplies both the MLR stream and the Clarifier. Clarifier underflow is split into RAS, which returns internally, and WAS, which leaves the plant.

Using Q_0 as the hydraulic basis makes the flow accounting visible before any kinetic equation is introduced:

$$q_P = 1 + r_I + r_R, \qquad q_C = 1 + r_R, \qquad q_U = r_R + w, \qquad q_E = 1 - w. \tag{1}$$

The Clarifier identity $q_C = q_E + q_U$ follows immediately. The operating domain requires $r_I \geq 0$, $r_R \geq 0$, $0 \leq w < 1$, and $q_U = r_R + w > 0$, so both outlet concentrations can later be recovered safely from their mass flows. At the mixer, fresh feed, MLR, and RAS meet. Dividing its component balance by Q_0 gives

$$q_P m = x + r_I c_N + r_R c_U. \tag{2}$$

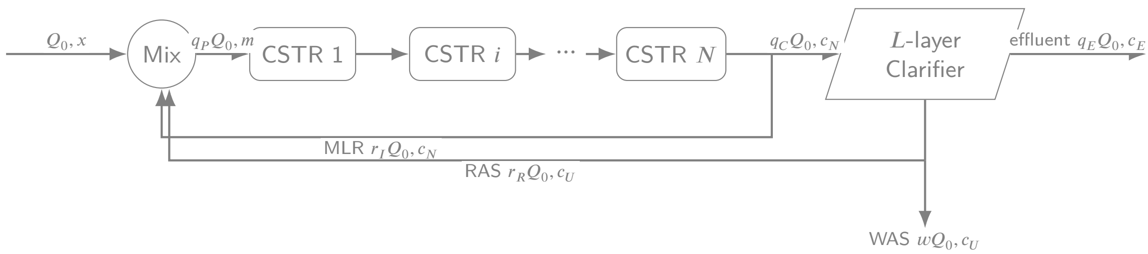

Figure 1: General recycling mixer–reactor–Clarifier system. Internal recycle streams change unit conditions but cancel from the external plant balance.

The residence time of each stage must be distinguished from the installed total. Define

$$h_i = \frac{24V_i}{Q_0}, \qquad H = \sum_{i=1}^{N} h_i, qquad d_i = \frac{Q_0 q_P}{V_i} = \frac{24 q_P}{h_i}, \tag{3}$$

where h_i and H are in hours and d_i is in d^{-1}. Thus an engineer may vary individual stage volumes through h_i, or impose fixed volume fractions $h_i = f_i^V H$ with $\sum_i f_i^V = 1$. Equal tanks are only the special choice $f_i^V = 1/N$.

Let $\vartheta \in \mathbb{R}^{d_\vartheta}$ contain exactly the quantities allowed to vary in a particular study. It may contain selected h_i, selected aeration settings a_i, stage-addition rates, and the three recycle/wasting ratios. Here a stage addition represented by a source term is understood to have negligible carrier flow. If its liquid flow is not negligible relative to Q_0, that stream must instead be included in the hydraulic balances, which changes the downstream q_P and d_i. Fixed settings are parameters rather than zero-variance regression inputs. This convention permits a different operating freedom at each stage without changing the derivation below.

2.2. One reactor balance, then the full train

Picture a control volume around reactor i. Material enters from the preceding stage, leaves at the same train flow, is transformed by biological and chemical processes, and may receive oxygen or a declared external addition. With $c_0 = m$, its steady component balance is

$$0 = d_i(c_{i-1} - c_i) + \nu^\top \rho_i(c_i; \vartheta) + \omega_i(a_i, c_i) e_{S_O} + \xi_i(\vartheta), \qquad i = 1, \dots, N. \tag{4}$$

Here $\nu \in \mathbb{R}^{n_p \times F}$ stores the stoichiometric change caused by n_p processes, $\rho_i \in \mathbb{R}_+^{n_p}$ contains their volumetric rates, e_{S_O} selects dissolved oxygen, $\omega_i = k_L a_i(a_i)(S_O^* - c_{i,S_O})$ is oxygen transfer, and ξ_i is any explicitly modeled negligible-flow stage addition in concentration-per-time units. Every term therefore has units of component concentration per day. Retaining ξ_i shows how a carbon or chemical dose must enter the balance if it is optimized. Any nonzero source must also satisfy the coordinate-face inward-pointing test used for the reaction model; otherwise non-negativity would not be dynamically invariant.

Some combinations of components are changed neither by the reaction network nor by oxygen transfer. Let $A \in \mathbb{R}^{K \times F}$ contain independent, physically identified inventory rows satisfying

$$A\nu^\top = 0, \qquad A e_{S_O} = 0, \qquad \operatorname{rank}(A) = K, \qquad \|A_{k,:}\|_2 = 1. \tag{5}$$

Premultiplication of Equation (4) by A removes the reaction and aeration terms and exposes what must pass from one tank to the next:

$$A(c_i - c_{i-1}) = d_i^{-1} A \xi_i(\vartheta), \qquad i = 1, \dots, N. \tag{6}$$

When $\xi_i = 0$, the quantity Ac_i is constant across the stage. These relations can be visualized as K inventory rails running through the train: components may exchange mass within a rail through reaction, but the rail itself cannot jump. Equation (6) also prevents an important modeling error: an optimized stage dose may retain a zero right-hand side only if $A\xi_i = 0$.

2.3. Clarifier bookkeeping before settling detail

At the Clarifier, component flow is more numerically useful than underflow concentration because q_U may be small. Define

$$g_E = q_E c_E, \qquad g_U = q_U c_U. \tag{7}$$

The component balance around the complete vessel is then

$$q_C c_N = g_E + g_U. \tag{8}$$

Because $q_E, q_U > 0$, no information is lost: $c_E = g_E/q_E$ and $c_U = g_U/q_U$.

Soluble material follows the liquid, whereas particulate material may be concentrated into the underflow. The linear relations retained in the projection are

$$E_S g_U = q_U E_S c_N, \qquad E_X g_U \geq q_U E_X c_N. \tag{9}$$

The first relation, together with Equation (8), also gives $E_S g_E = q_E E_S c_N$. The second says that the underflow particulate concentration cannot be lower than the feed concentration. With non-negative outlet flows it bounds each particulate recovery between q_U/q_C and one.

Let s_ℓ be the TSS concentration in layer ℓ, numbered from the water surface ($\ell = 1$) to the hopper ($\ell = L$). The endpoint layers must agree with the TSS carried by the outlet component flows:

$$q_E s_1 = t_X^\top g_E, \qquad q_U s_L = t_X^\top g_U. \tag{10}$$

The convex envelope

$$s_1 \leq s_\ell \leq s_L, \qquad \ell = 2, \ldots, L-1, \tag{11}$$

keeps every internal layer between the two outlet concentrations. It is intentionally an endpoint envelope, not a claim of layer-by-layer monotonicity. The nonlinear settling-flux balances, which determine the detailed shape, remain part of the mechanistic model. Once the layer values are known, their physical inventory is not an independent variable but

$$M_{\mathrm{cl}} = \sum_{\ell=1}^{L} V_{\mathrm{cl},\ell} s_\ell. \tag{12}$$

2.4. Raw complete-system surrogate

The regression must predict every quantity needed to close the recycle and evaluate the plant. Stack the outputs in process order:

$$\chi = \mathrm{col}(m, c_1, \ldots, c_N, g_E, g_U, s_1, \ldots, s_L) \in \mathbb{R}^{n_\chi}, \qquad n_\chi = (N+3)F + L. \tag{13}$$

This ordering lets a reader move through the vector exactly as water moves through Figure 1.

Let $z = \mathrm{col}[D_{\vartheta,\mathrm{fit}}^{-1}(\vartheta - \bar{\vartheta}), D_{x,\mathrm{fit}}^{-1}(x - \bar{x})] \in \mathbb{R}^{d_z}$, where $d_z = d_\vartheta + F$ and bars and positive diagonal fitting scales are computed from the development data. To admit curvature and interactions without duplicating terms, form

$$r(z) = \mathrm{col}[z, \mathrm{vech}(zz^\top)], \qquad \phi(\vartheta, x) = \mathrm{col}[1, D_{r,\mathrm{fit}}^{-1}\{r(z) - \bar{r}\}] \in \mathbb{R}^{n_\phi}, \tag{14}$$

where vech lists one triangular copy of the symmetric quadratic matrix. Hence

$$n_\phi = 1 + d_z + \frac{d_z(d_z+1)}{2}. \tag{15}$$

The dimension now adapts automatically if another stage control is freed.

With development response mean $\bar{\chi}$, positive scale D_χ, and fitted coefficient matrix $\widehat{C}_{\gamma_{\mathrm{R}}}$, the raw response is

$$\chi_{\mathrm{raw}}(\vartheta, x) = \bar{\chi} + D_\chi \widehat{C}_{\gamma_{\mathrm{R}}} \phi(\vartheta, x). \tag{16}$$

The ridge parameter γ_{R} controls statistical complexity. At this point Equation (16) is only a regression prediction; no sign or balance property has yet been imposed.

2.5. From a raw prediction to one physically admissible response

The projection is built by collecting the physical statements already derived, rather than introducing unrelated constraints at once. Define $\delta_i(\vartheta) = d_i^{-1} A \xi_i(\vartheta)$ and let $\mathcal{H}(\vartheta)\chi = \mathfrak{b}(\vartheta, x)$ represent

$$\operatorname{col}\begin{pmatrix} q_P m - r_I c_N - (r_R/q_U) g_U - x \\ \{A(c_i - c_{i-1}) - \delta_i\}_{i=1}^N, \quad c_0 = m \\ g_E + g_U - q_C c_N \\ E_S g_U - q_U E_S c_N \\ q_E s_1 - t_X^\top g_E \\ q_U s_L - t_X^\top g_U \end{pmatrix} = 0. \tag{17}$$

There are

$$n_h = 2F + NK + F_S + 2 \tag{18}$$

rows: one mixer balance, N invariant transitions, one full Clarifier balance, soluble pass-through, and two TSS endpoint identities.

The remaining linear physical directions are written $\mathcal{G}(\vartheta)\chi \le 0$:

$$\mathcal{G}(\vartheta)\chi = \operatorname{col}\big(q_U E_X c_N - E_X g_U, s_1 \mathbf{1}_{L-2} - s_{2:L-1}, s_{2:L-1} - s_L \mathbf{1}_{L-2}\big) \le 0. \tag{19}$$

Thus $n_g = F_X + 2(L-2)$ inequality rows describe particulate densification and the layer envelope. Non-negativity contributes another n_χ inequalities.

The correction is measured in standardized response coordinates so that a large-magnitude TSS coordinate does not overwhelm a small nutrient coordinate. The projected displacement and response are

$$\begin{aligned} \widehat{u}(\vartheta, x) = \underset{u \in \mathbb{R}^{n_\chi}}{\arg\min} \quad & \tfrac{1}{2}\|u\|_2^2 && (20)\\ \text{subject to} \quad & D_b^{-1}\{\mathcal{H}(\vartheta)[\chi_{\text{raw}} + D_\chi u] - \mathfrak{b}(\vartheta, x)\} = 0, \\ & \chi_{\text{raw}} + D_\chi u \ge 0, \\ & D_g^{-1}\mathcal{G}(\vartheta)[\chi_{\text{raw}} + D_\chi u] \le 0, \\ \widehat{\chi}(\vartheta, x) = \chi_{\text{raw}}(\vartheta, x) &+ D_\chi \widehat{u}(\vartheta, x). && (21) \end{aligned}$$

D_b and D_g improve numerical conditioning but do not alter the feasible set. Whenever that set is nonempty, the identity Hessian makes the objective strictly convex and coercive, so the projected response exists and is unique. In the no-addition case, nonemptiness holds at every declared design point by construction: set $m = c_i = x$ for all i, $g_E = q_E x$, $g_U = q_U x$, and $s_\ell = t_X^\top x$ for every layer. For nonzero additions, nonemptiness is instead a required domain-admissibility check because the invariant right-hand sides change.

The projection has a useful audit property. If a mechanistic reference χ_{ref} satisfies the projection constraints exactly, Euclidean-projection geometry in standardized coordinates gives

$$\|D_\chi^{-1}(\widehat{\chi} - \chi_{\text{ref}})\|_2^2 \le \|D_\chi^{-1}(\chi_{\text{raw}} - \chi_{\text{ref}})\|_2^2 - \|\widehat{u}\|_2^2. \tag{22}$$

Thus the physical correction cannot increase standardized squared error to a reference point that is actually in the projection set. A numerically near-feasible target requires the finite-distance bound introduced in the methodology; small infeasibility is not treated as exact feasibility. Neither result proves that the projected state satisfies all kinetic or nonlinear settling equations, which are tested separately.

2.6. Engineering objective and operating constraints

The optimizer needs a scalar objective, but the priority assigned to each engineering concern is a case-study choice rather than a universal constant. Let $z_E = I_{\text{comp}} c_E \in \mathbb{R}_+^4$ contain effluent COD, TN, TP, and TSS. With positive normalization D_T and quality priorities $\varpi_Q \ge 0$, $\mathbf{1}^\top \varpi_Q = 1$, define

$$j_Q = \varpi_Q^\top D_T^{-1} z_E. \tag{23}$$

Representative dimensionless resource terms are

$$
\begin{aligned}
j_H &= \frac{H - H^L}{H^U - H^L}, & j_A &= \frac{\sum_{i\in\mathcal{A}} V_i k_L a_i}{E_A^{\mathrm{ref}}}, & & \\
j_I &= \frac{r_I - r_I^L}{r_I^U - r_I^L}, & j_R &= \frac{r_R - r_R^L}{r_R^U - r_R^L}, & j_W &= \frac{w\, t_X^\top c_U}{w^U X_U^{\mathrm{ref}}}.
\end{aligned} \tag{24}
$$

$\mathcal{A}$ is the set of stages whose aeration can be nonzero, and $E_A^{\mathrm{ref}} > 0$ is a fixed normalization chosen before optimization; for the case study it is the maximum value of $\sum_{i\in\mathcal{A}} V_i k_L a_i$ over the declared operating box. The aeration term is therefore proportional to total installed oxygen-transfer capacity and does not acquire a moving denominator when stage volumes are decisions. Additional declared operating terms can be appended in the same manner. With $j = \mathrm{col}(j_Q, j_H, j_A, j_I, j_R, j_W)$, the objective is

$$
J(\vartheta, \chi; \varpi) = \varpi^\top j(\vartheta, \chi), \qquad \varpi \geq 0, qquad \mathbf{1}^\top \varpi = 1. \tag{25}
$$

The symbols ϖ and ϖ_Q deliberately expose both levels of priority; their numerical values are stated and justified in the case study.

Several familiar plant constraints can be calculated from the same projected response. Define $X_i = t_X^\top c_i$, $X_E = t_X^\top c_E$, $X_U = t_X^\top c_U$, and the external solids loss rate

$$
\dot{M}_X = Q_0(q_E X_E + w X_U). \tag{26}
$$

The whole-plant solids retention time, surface overflow rate, and Clarifier solids loading rate are then

$$
\mathrm{SRT} = \frac{\sum_{i=1}^N V_i X_i + M_{\mathrm{cl}}}{\dot{M}_X}, \qquad \dot{M}_X \geq \epsilon_X > 0, \tag{27}
$$

$$
\mathrm{SOR} = \frac{Q_0 q_E}{A_{\mathrm{cl}}}, \tag{28}
$$

$$
\mathrm{SLR} = \frac{10^{-3} Q_0 q_C X_N}{A_{\mathrm{cl}}}. \tag{29}
$$

The factor 10^{-3} converts grams to kilograms in SLR. Symbolic lower and upper limits on SRT, SOR, SLR, X_U, and any effluent targets form $g_{\mathrm{eng}}(\vartheta, \chi) \leq 0$; only the case-study section assigns their values. In computation the SRT bounds are imposed in cross-multiplied form after the positive-loss guard, avoiding division by a small quantity.

Physical feasibility alone does not prevent optimization from asking the projection to make an unusually large repair or from exploiting a state that conserves inventories but is kinetically implausible. Five development-supported diagnostics therefore define $g_{\mathrm{trust}} \leq 0$: standardized correction $\kappa_{\mathrm{corr}} = \|\hat{u}\|_2/\sqrt{n_\chi}$; regularized feature leverage; deviation from a common particulate recovery; the scaled smooth reactor-balance residual; and the scaled smooth Clarifier-layer residual. Their explicit formulas and thresholds are provided in the Supplementary Material. Each threshold is determined without using the untouched test or optimization responses.

The surrogate-based operational problem is

$$
\begin{aligned}
\min_{\vartheta} \quad & J[\vartheta, \hat{\chi}(\vartheta, x); \varpi] \\
\text{subject to} \quad & \vartheta \in \mathcal{V}, \\
& g_{\mathrm{eng}}[\vartheta, \hat{\chi}(\vartheta, x)] \leq 0, \\
& g_{\mathrm{trust}}[\vartheta, \chi_{\mathrm{raw}}, \hat{\chi}] \leq 0, \\
& \hat{\chi}(\vartheta, x) \text{ solves Equation (20)}.
\end{aligned} \tag{30}
$$

Here $\mathcal{V}$ contains the declared bounds and any stage-allocation equalities. The lower response is unique; the upper problem remains generally nonconvex and piecewise smooth. Accordingly, ϑ_S denotes the selected, independently verified local incumbent rather than a certified global minimizer.

2.7. An exact lower-level certificate without complementarity slacks

For fixed (ϑ, x), write the projection as

$$\min_{u} \tfrac{1}{2}u^\top u \quad \text{subject to} \quad \mathsf{E}u = e, \qquad \mathsf{G}_u u \le g. \tag{31}$$

The dimensions are $u \in \mathbb{R}^{n_\chi}$, $\mathsf{E} \in \mathbb{R}^{n_h \times n_\chi}$, and $\mathsf{G}_u \in \mathbb{R}^{n_q \times n_\chi}$ with $n_q = n_\chi + n_g$. Let $\lambda^=$ be the unrestricted equality multiplier and $\lambda^\le \ge 0$ the inequality multiplier. The dual value at any dual-feasible pair is

$$d_{\mathrm{QP}}(\lambda^=, \lambda^\le) = -\tfrac{1}{2}\|\mathsf{E}^\top \lambda^= + \mathsf{G}_u^\top \lambda^\le\|_2^2 - e^\top \lambda^= - g^\top \lambda^\le. \tag{32}$$

For a primal-feasible u, weak duality makes the primal–dual gap non-negative. Using $\mathsf{E}u = e$, the gap can be written as the sum of two non-negative quantities:

$$\begin{aligned}\Gamma &= \tfrac{1}{2}u^\top u - d_{\mathrm{QP}} \\ &= \tfrac{1}{2}\|u + \mathsf{E}^\top \lambda^= + \mathsf{G}_u^\top \lambda^\le\|_2^2 + (\lambda^\le)^\top (g - \mathsf{G}_u u) \ge 0.\end{aligned} \tag{33}$$

Consequently, $\Gamma = 0$ is equivalent to stationarity and complementarity and certifies that u is the unique lower-QP solution. This strong-duality construction removes the n_q explicit slack variables used by a conventional complementarity lift (Boyd and Vandenberghe, 2004).

The single-level calculation replaces the last line of Equation (30) by primal feasibility, $\lambda^\le \ge 0$, and the dimension-normalized condition $\overline{\Gamma} = \Gamma/(n_\chi + n_q) \le \tau$. A decreasing positive sequence of τ values is used for numerical continuation. The final controls are accepted only after the original QP is solved independently with a convex-QP method (Stellato, Banjac, Goulart, Bemporad, and Boyd, 2020) and its primal feasibility, dual feasibility, stationarity, complementarity, and upper-level constraints all pass their stated tolerances. Multiple deterministic starts seek distinct local basins; they are not a global-optimality certificate.

2.8. Direct smooth mechanistic comparator

The comparison route removes both regression and projection. Its state contains the concentrations that actually accumulate in the reduced model:

$$y = \mathrm{col}(c_1, \dots, c_N, s_1, \dots, s_L) \in \mathbb{R}_+^{n_y}, \qquad n_y = NF + L. \tag{34}$$

The mixer and Clarifier outlet vectors are reconstructed algebraically from (ϑ, x, y). To keep the common-composition division defined even at infeasible solver iterates, introduce an auxiliary Clarifier-feed TSS $\widetilde{X}_F \ge X_F^{\min} > 0$ and impose $\widetilde{X}_F = t_X^\top c_N$; reconstruction uses $\widetilde{X}_F$. A differentiable residual $f_{\mathrm{sm}}(\vartheta, x, y, \widetilde{X}_F)$ contains all NF reactor balances and all L finite-volume Clarifier balances. To ensure that both optimization routes use the same admissible layer ordering, define $g_{\mathrm{env}}(y) = \mathrm{col}(s_1 \mathbf{1}_{L-2} - s_{2:L-1}, s_{2:L-1} - s_L \mathbf{1}_{L-2})$. In this route the auxiliary lower bound already imposes the feed-TSS engineering limit, so let $g_{\mathrm{eng,red}}$ denote g_{eng} with that duplicate row omitted. The direct problem is

$$\begin{aligned}\min_{\vartheta, y, \widetilde{X}_F} \quad & J[\vartheta, \chi(\vartheta, x, y, \widetilde{X}_F); \varpi] \\ \text{subject to} \quad & f_{\mathrm{sm}}(\vartheta, x, y, \widetilde{X}_F) = 0, \\ & \widetilde{X}_F = t_X^\top c_N, \quad \widetilde{X}_F \ge X_F^{\min}, \\ & \vartheta \in \mathcal{V}, \quad y \ge 0, \\ & g_{\mathrm{env}}(y) \le 0, \\ & g_{\mathrm{eng,red}}[\vartheta, \chi(\vartheta, x, y, \widetilde{X}_F)] \le 0.\end{aligned} \tag{35}$$

Its selected, independently verified local incumbent is denoted $(\vartheta_M, y_M, \widetilde{X}_{F,M})$. The auxiliary is algebraically redundant at feasibility but makes every trial-point denominator safe. Smooth approximations are used only for the minimum, maximum, positive-part, zero-denominator, and receiver-switch operations in the settling and kinetic equations. A nonzero smoothing width cannot be globally identical to a nonsmooth function. The smooth model is

Table 2
Case-study topology, geometry, and free-coordinate bounds.

Quantity	Value or range	Case-study role
Q_0	10,000 m^3 d^{-1}	Fresh-flow basis
N; L; ℓ_F	5; 10; 5	Reactors; layers; feed layer
f_i^V	$1/5$	Equal reactor-volume fractions
A_{cl}; depth	1500 m^2; 4.0 m	Clarifier geometry
H	6–36 h	Total nominal HRT
a_3, a_4, a_5	independently 0–1	Stage aeration settings
r_I	0–4	MLR ratio
r_R	0.25–1.25	RAS ratio
w	0.001–0.05	WAS ratio

therefore called numerically equivalent only on points that pass predeclared state, objective, constraint, residual, and branch comparisons against the original nonsmooth reduced reference equations.

The scalar-layer Clarifier is an aggregate-solids steady-state model. Particulate outlet components are reconstructed with the steady feed composition; consequently, a BDF march is a pseudo-transient numerical route to the nonsmooth steady equations, not a claim that every particulate species is dynamically inventoried inside every layer. A reference replay means an independently converged solution of those original, nonsmoothed equations. Stability refers only to the reduced numerical state model.

3. Methodology

3.1. Case-study definition

All principal case topology, decision bounds, objective priorities, and engineering limits are collected here so they cannot be mistaken for requirements of the general formulation; kinetic, settling, and numerical constants are tabulated in the Supplementary Material. The case has $F = 20$ components from ASM2d with two-step nitrification (ASM2d-TSN), $F_S = F_X = 10$, and $K = 5$ named invariants. Five equal-volume reactors are followed by a ten-layer Clarifier. Reactors 1 and 2 are unaerated; reactors 3–5 have independent aeration controls. Thus

$$\vartheta = (H, a_3, a_4, a_5, r_I, r_R, w)^\top \in \mathbb{R}^7, \tag{36}$$

while $h_i = H/5$, $a_1 = a_2 = 0$, $k_L a_i = 47 a_i$ d^{-1} for $i = 3, 4, 5$, and $\xi_i = 0$ for every stage.

Five equal-volume stages provide a compact anoxic–aerobic train for the comparative calculation, while ten 0.4-m Clarifier layers resolve the specified 4-m depth without making the layer count a structural constant. The operating box deliberately spans low-to-high residence time, aeration, recycle, and wasting settings while preserving $q_E > 0$ and $q_U > 0$. These choices define the case-study domain; they are not asserted as universal design bounds.

For this case, $d_z = d_\vartheta + F = 27$, $n_\phi = 406$, $n_\chi = 170$, $n_y = 110$, $n_h = 77$, $n_g = 26$, and $n_q = 196$. These are evaluations of Equations (13), (15), (18), and (19); they are not fixed architectural constants. A different N, L, or free-control set requires a correspondingly dimensioned design and a newly fitted surrogate.

The overall objective priorities are

$$\varpi = (0.50, 0.15, 0.20, 0.05, 0.05, 0.05)^\top, \qquad \varpi_Q = \tfrac{1}{4}\mathbf{1}_4. \tag{37}$$

The quality coefficient is the largest single trade-off coefficient; aeration receives the largest resource coefficient because it is ordinarily the dominant manipulated energy demand; HRT represents installed biological volume; and the remaining coefficients are shared among MLR, RAS, and wasted-solids production. Because the normalized objective components are not all confined to the same interval, these coefficients are trade-off parameters rather than percentages of the attained objective. These values define this comparative case only. D_T contains the positive development-set standard deviations of effluent COD, TN, TP, and TSS. The fixed aeration normalization is $E_A^{\mathrm{ref}} = \sum_{i=3}^{5} V_i(H^U)(47\,\mathrm{d}^{-1}) = 423{,}000$ m^3 d^{-1}, the maximum transfer-capacity proxy in the case box.

The case imposes $8 \le \mathrm{SRT} \le 30$ d, $\mathrm{SLR} \le 100$ kg TSS m^{-2} d^{-1}, $X_U \le 15{,}000$ g TSS m^{-3}, $X_N \ge 1$ g TSS m^{-3}, and $\dot{M}_X/Q_0 \ge 1$ g TSS m^{-3}. The wasted-solids objective uses $X_U^{\mathrm{ref}} = 15{,}000$ g TSS m^{-3}. SOR is reported but not passed to the optimizer: the declared Q_0, A_{cl}, and w range imply $6.33 \le \mathrm{SOR} \le 6.66$ m d^{-1}, so a 20 m d^{-1} upper row would be provably redundant. These are case choices, not universal activated-sludge limits. Influent ranges, Clarifier settling constants, kinetic parameters, and all numerical tolerances are listed in the Supplementary Material.

3.2. Study sequence and protection against information leakage

The calculation follows six frozen phases:

1. generate 1,000 accepted mechanistic steady states over the joint seven-control and twenty-influent domain;
2. place 800 states in the development block and 200 independently randomized states in an untouched test block;
3. select the ridge parameter by five-fold cross-validation entirely within the 800 development states and fit one final development surrogate;
4. evaluate raw and projected responses once on the untouched 200 states;
5. solve the nominal and ten influent scenarios by both Equations (30) and (35); and
6. replay both selected decisions with the original nonsmooth steady-state model.

Test states and optimization responses never determine feature definitions, scales, ridge selection, objective priorities, limits, smoothing widths, tolerances, or starts.

The ordered component vector is

$$\begin{aligned}&(S_O, S_F, S_A, S_{NH4}, S_{NO2}, S_{NO3}, S_{N2}, S_{PO4}, S_I, S_{ALK},\\ &\ X_I, X_S, X_H, X_{PAO}, X_{PP}, X_{PHA}, X_{AOB}, X_{NOB}, X_{MeP}, X_{MeOH}).\end{aligned} \tag{38}$$

Independent randomized Latin-hypercube blocks supply the 800 development and 200 test inputs in 27 dimensions (McKay, Beckman, and Conover, 1979). Rows within a block are stratified and are not described as independent observations. A failed or unstable mechanistic point is not silently replaced; an unresolved design point stops the study so solver success cannot condition the intended design.

3.3. Five-fold cross-validation and final ridge fit

The 800 development rows are partitioned reproducibly into five folds of 160. For a fitting index set $\mathcal{I}$, let $\Phi_{\mathcal{I}}$ contain feature rows, $\widetilde{Y}_{\mathcal{I}}$ contain standardized responses, and $R_\phi = \mathrm{Diag}(0, 1, \dots, 1)$ leave the intercept unpenalized. Training solves

$$\widehat{C}_{\gamma_{\mathrm{R}}} = \arg\min_{C} \left\{ \frac{\|\widetilde{Y}_{\mathcal{I}} - \Phi_{\mathcal{I}} C^\top\|_F^2}{|\mathcal{I}| n_\chi} + \frac{\gamma_{\mathrm{R}}}{n_\chi} \|C R_\phi\|_F^2 \right\}. \tag{39}$$

Every center and scale is recomputed from the 640 fitting rows of a fold. Positive candidate values of γ_{R} guarantee a unique, stable coefficient solution even if a feature matrix is nearly rank deficient. The fold score is complete-state standardized RMSE of the raw response. The one-standard-error rule chooses the largest γ_{R} whose mean score is within one standard error of the minimum (Hastie, Tibshirani, and Friedman, 2009). Projection metrics are recorded but do not select γ_{R}, so reconciliation cannot hide a weak raw model. The chosen value is refitted once on all 800 development rows.

The untouched test reports RMSE, MAE, bias, normalized RMSE, and R^2 for the mixer, each reactor, both outlet-flow blocks, every Clarifier layer, and the complete response. Raw and projected predictions are reported side by side. For test row i, let C_i be its projection polyhedron and let P_{C_i} be metric projection in the norm $\|v\|_{D_\chi^{-1}} := \|D_\chi^{-1} v\|_2$. Because a numerical mechanistic target χ_i satisfies equations only to finite tolerance, its feasibility distance $\delta_{\mathrm{feas},i} = \|P_{C_i}(\chi_i) - \chi_i\|_{D_\chi^{-1}}$ is reported. The following general bound is checked for every target:

$$\|P_{C_i}(\chi_{\mathrm{raw},i}) - \chi_i\|_{D_\chi^{-1}} \le \|\chi_{\mathrm{raw},i} - \chi_i\|_{D_\chi^{-1}} + \delta_{\mathrm{feas},i}. \tag{40}$$

Equation (22) is invoked only for a target known to be exactly feasible; a merely small positive $\delta_{\mathrm{feas},i}$ does not justify the stronger squared inequality. Optimization begins only if all QPs pass their primal–dual audit, complete-state raw nRMSE is below the predeclared case threshold, and every development-supported trust threshold has been frozen successfully. A failed gate ends the study without refitting.

3.4. Nominal optimization, mechanistic comparison, and reference replay

The nominal influent is the midpoint of each declared influent interval. Both optimization routes use the same free controls, bounds, objective priorities, engineering constraints, and nine deterministic starts in the normalized seven-dimensional box. For the surrogate route, each continuation stage uses the primal–dual-gap tolerance sequence $10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}$. Its final operating vector is accepted only after an independently initialized projection QP reproduces the projected state and all unrelaxed upper constraints.

After continuation, an outer refinement solves the original projection QP to its acceptance tolerances at every trial point. On a stable active set, derivatives follow from the lower-QP KKT sensitivity system; an active-set change triggers a fresh QP and derivative reconstruction. A surrogate-route candidate is called a first-order KKT-stationary feasible point only if its independently recomputed scaled primal, dual, stationarity, and complementarity residuals pass the declared tolerances. The direct mechanistic route is subjected to the analogous full-space KKT audit, including every mechanistic equality and the auxiliary feed-solids equality. If no stationary candidate is found, the best validated feasible point is reported more conservatively as a stationarity-unresolved local incumbent. First-order stationarity is not a certificate of local minimality.

For $k \in \{S, M\}$, let $\chi_{\mathrm{ref}}(\vartheta_k, x)$ be the reference nonsmooth replay at the surrogate-based or mechanistic-NLP decision. Raw, projected, and smooth responses are all evaluated at both decisions so prediction error is not confounded with decision difference:

$$
\begin{aligned}
e_{\mathrm{raw,k}} &= D_\chi^{-1}[\chi_{\mathrm{raw}}(\vartheta_k, x) - \chi_{\mathrm{ref}}(\vartheta_k, x)],\\
e_{\mathrm{proj,k}} &= D_\chi^{-1}[\hat{\chi}(\vartheta_k, x) - \chi_{\mathrm{ref}}(\vartheta_k, x)],\\
e_{\mathrm{sm,k}} &= D_\chi^{-1}[\chi_{\mathrm{sm}}(\vartheta_k, x) - \chi_{\mathrm{ref}}(\vartheta_k, x)],\\
\Delta J_{\mathrm{pred,S}} &= J(\vartheta_S, \hat{\chi}_S) - J(\vartheta_S, \chi_{\mathrm{ref,S}}),\\
\Delta J_{S-M} &= J(\vartheta_S, \chi_{\mathrm{ref,S}}) - J(\vartheta_M, \chi_{\mathrm{ref,M}}).
\end{aligned}
\tag{41}
$$

ΔJ_{S-M} compares two verified local incumbents; it is not called regret because neither route supplies a global certificate.

The smooth mechanistic equations are also benchmarked at all 200 untouched test inputs and at every available selected decision—at most 22 decisions from two routes across eleven optimization cases. The actual selected-decision denominator is reported; a failed route contributes no fictitious decision. At each fixed input, both the smooth and reference equations are solved independently from the two declared influent-based starts; the optimized state is not used to initialize this comparison. Let D_y be the positive diagonal mechanistic-state scale defined in the Supplementary Material. With $n_y = 110$ in the case study, acceptance requires

$$
\frac{\|D_y^{-1}(y_{\mathrm{sm}} - y_{\mathrm{ref}})\|_2}{\sqrt{n_y}} \le 10^{-6}, \qquad \|D_y^{-1}(y_{\mathrm{sm}} - y_{\mathrm{ref}})\|_\infty \le 10^{-5},
\tag{42}
$$

relative objective and engineering-quantity differences no greater than 10^{-6}, own-model scaled residuals no greater than 10^{-8}, a cross-model scaled residual no greater than 10^{-5}, identical feasibility classifications, and matching receiver, storage-capacity, settling-floor, settling-cap, and flux-minimum branches. At a selected mechanistic decision, the independent fixed-input smooth state must also reproduce the optimizer's reported state. Reduced-state stability requires the rightmost computed eigenvalue to lie below a strictly negative margin after a finite-difference error allowance. Passing these finite-domain tests supports only the phrase "numerically equivalent on the declared study domain."

3.5. Ten-case influent sensitivity scenarios

Ten new influent vectors are drawn by an independent 20-dimensional Latin hypercube over the stated influent ranges. No response or decision from these cases enters training or tuning. Both optimization routes are attempted and adjudicated for each case; reference replay is attempted wherever a route supplies a selected decision. The analysis reports each route's status and, where available, all seven selected controls; stagewise dissolved oxygen, TN, TP, and TSS; Clarifier outlet composites and layers; SRT, SOR, and SLR; objective components; projection correction; and solve time. Ten cases support descriptive casewise comparisons, not population-level reliability claims.

Table 3
Five-fold selection and untouched test performance. Raw and projected errors are required for every block.

Block	CV raw	Test raw	Test projected	Correction RMS
Mixer	–	–	–	–
Reactors 1–N (each reported)	–	–	–	–
Clarifier overflow	–	–	–	–
Clarifier underflow	–	–	–	–
Clarifier layers	–	–	–	–
Complete response	–	–	–	–

Table 4
Required trust-region audit. Test summaries are descriptive; the frozen limits constrain surrogate optimization.

Diagnostic	Frozen limit	Test 95th percentile	Largest selected value
Projection correction κ_{corr}	–	–	–
Regularized leverage ℓ_{R}	–	–	–
Particulate split κ_{split}	–	–	–
Reactor residual κ_{rxn}	–	–	–
Clarifier flux κ_{flux}	–	–	–

Table 5
Nominal selected controls and reference-replay performance.

Quantity	Surrogate-based solution	Direct mechanistic solution
$H, a_3, a_4, a_5, r_I, r_R, w$	–	–
Projected-surrogate objective $J(\vartheta_S, \hat{\chi}_S)$	–	n.a.
Smooth-mechanistic objective $J(\vartheta_M, \chi_{\mathrm{sm},M})$	n.a.	–
Reference-replay objective	–	–
Reference COD, TN, TP, TSS	–	–
Reference SRT and underflow TSS	–	–
Total solution time	–	–

3.6. Timing and numerical acceptance

Timing separates raw regression, one projection QP, complete surrogate optimization, smooth mechanistic optimization, and reference steady-state replay. The same processor allocation and thread count are used for the two optimization routes. Setup and solve times, attempted and accepted starts, iterations, QP resolutions, and replays are all reported. A solver success flag is never sufficient: scaled primal, dual, stationarity, complementarity, physical-balance, reduced-stability, domain, and branch checks must also pass the Supplementary Material contract.

4. Results and discussion

No numerical result is inserted before the complete 1,000-state design and all eleven paired optimization cases have been attempted and adjudicated under the frozen checks. Failed or unresolved cases remain visible in the denominators and result tables rather than being replaced or omitted. Tables 3–12 define the minimum result set and prevent selective reporting after outcomes are known.

4.1. Cross-validation and untouched prediction performance

Discussion will distinguish native regression accuracy from improvement caused by convex reconciliation. Exact balances do not by themselves imply kinetic accuracy, so error, correction, nonlinear-residual, and support diagnostics are interpreted together.

Table 6
Required nominal process-profile comparison. The final table contains separate panels at ϑ_S and ϑ_M and expands every reactor and Clarifier layer.

Location and reported quantity	Raw	Projected	Smooth	Reference
Mixer: COD, TN, TP, TSS	–	–	–	–
Reactor $i = 1, \dots, N$: DO, TN, TP, TSS	–	–	–	–
Clarifier overflow: COD, TN, TP, TSS	–	–	–	–
Clarifier underflow: COD, TN, TP, TSS	–	–	–	–
Layer $\ell = 1, \dots, L$: TSS	–	–	–	–

Table 7
Smooth-NLP versus reference nonsmooth comparison over the 200 test states and every available selected decision (at most 22); the actual denominator is reported.

Comparison	Median	95th percentile	Maximum
Available selected decisions, n_{sel}		–	
Total evaluated fixed-input points, $200 + n_{\mathrm{sel}}$		–	
Smooth two-start scaled disagreement	–	–	–
Reference two-start scaled disagreement	–	–	–
Optimizer-state reproduction error	–	–	–
Scaled state RMS difference	–	–	–
Scaled state infinity difference	–	–	–
Own-model scaled residual	–	–	–
Cross-model scaled residual	–	–	–
Relative objective difference	–	–	–
Maximum engineering-row difference	–	–	–
Rightmost reduced-state eigenvalue, d^{-1}	–	–	–
Feasibility-class mismatches		–	
Branch-classification mismatches		–	

Table 8
Required casewise influent-scenario comparison. The full table contains ten rows. Here $\mathrm{RMS}_{\mathrm{proj,S}} = \|e_{\mathrm{proj,S}}\|_2/\sqrt{n_\chi}$, $\mathrm{RMS}_{\mathrm{sm,M}} = \|e_{\mathrm{sm,M}}\|_2/\sqrt{n_\chi}$, and the time ratio is complete mechanistic-route time divided by complete surrogate-route time.

Case	$J_{S,\mathrm{ref}}$	$J_{M,\mathrm{ref}}$	ΔJ_{S-M}	$\mathrm{RMS}_{\mathrm{proj,S}}$	$\mathrm{RMS}_{\mathrm{sm,M}}$	t_M/t_S
1–10	–	–	–	–	–	–

4.2. Nominal case and stagewise physical interpretation

The nominal discussion will trace raw, projected, smooth, and reference profiles from the mixer through each reactor and down the Clarifier. It will identify which inventories required correction, which controls or engineering constraints were active, and whether the two reference objectives differ materially.

4.3. Smooth mechanistic equivalence

Finite-domain numerical equivalence is claimed only when Table 7 satisfies the state limits in Equation (42) and the complete two-start, own-residual, cross-residual, objective, engineering, feasibility, optimizer-state, stability, and branch contract in the Supplementary Material.

4.4. Influent scenarios and optimization comparison

Casewise reporting is retained because ten scenarios are too few for distributional claims. Control movement is related to influent COD, nitrogen, phosphorus, and solids loading, and prediction error is separated from decision difference.

Table 9
Required selected controls for every influent case and optimization route. Each of the ten cases has one surrogate-route row and one mechanistic-route row.

Case	Route	H	a_3	a_4	a_5	r_I	r_R	w
1–10	S and M	–	–	–	–	–	–	–

Table 10
Required casewise water-quality comparison. At both selected decisions, predictor rows are raw, projected, smooth, and reference wherever defined.

Case	Decision	Response	COD	TN	TP	TSS	J
1–10	ϑ_S, ϑ_M	raw/projected/smooth/reference	–	–	–	–	–

Table 11
Required case-adjudication record. Every nominal and influent-scenario case remains in the denominator even when a route fails or is unresolved.

Case	Surrogate route	Mechanistic route	First applicable outcome
Nominal	–	–	–
Scenarios 1–10 (each reported)	–	–	–

Table 12
Timing and solver workload for the two optimization routes.

Metric	Surrogate plus projection	Smooth mechanistic NLP
Median start time	–	–
Median complete-case time	–	–
95th-percentile complete-case time	–	–
Accepted starts / attempted starts	–	–
Reference validation time	–	–

4.5. Computational comparison

The one-time costs of mechanistic data generation and fitting are reported separately from repeated optimization cost.

5. Conclusions

The formulation represents an N-stage recycling mixer–reactor–Clarifier plant with a raw statistical response and a unique physical projection. Its dimensions follow directly from N, L, the component partition, and the freely varied controls. Stagewise aeration and residence-time decisions are therefore supported without treating a five-stage arrangement as universal.

The projection has a deliberately limited role: it enforces the declared linear network physics and non-negativity while making the smallest standardized change to the raw prediction. The primal–dual-gap reformulation retains the exact lower-QP certificate at zero gap while avoiding a separate complementarity slack for every inequality. Nonlinear kinetics and settling are verified through reference nonsmooth replay rather than being silently credited to the projection. Numerical conclusions will be finalized only after all predeclared tables are populated without changing the model or analysis plan.

CRediT authorship contribution statement

The sole author, Egberto F. Selerio Jr., contributed to conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft, writing—review & editing, visualization, supervision, project administration, and funding acquisition.

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Data availability

The study record will contain the 1,000 mechanistic input–response states, immutable development/test and fold membership, fitted coefficients and scales, raw and projected test predictions, every nominal and influent-scenario optimization record, reference validation states, and timing summaries.

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